

Milestone 4 – Forecast Integration & Capacity Planning

1. Introduction

Milestone 4 focuses on deploying the trained machine learning model. It converts the model from development stage to practical usage. It is used for demand forecasting and capacity planning and helps in real-world decision-making.

2. Objectives

- Deploy trained ML model
- Generate predictions using data
- Store forecast results
- Evaluate model performance
- Simulate automation process
- Enable monitoring and retraining

3. Dataset Used

Economic dataset used for forecasting including:

GDP growth rate, Inflation rate, Unemployment rate, Tax burden, Population.

Data Processing:

- Cleaned missing values
- Structured dataset properly
- Applied feature engineering
- Created lag features for better prediction

4. Implementation (Step-by-Step)

Step 1: Data Loading – Dataset loaded using Python (Pandas)

Step 2: Data Preprocessing – Selected features, removed unnecessary columns, handled missing values

Step 3: Model Training – Used Random Forest, split data, trained model

Step 4: Model Deployment – Saved as forecast_model.pkl

Step 5: Prediction – Loaded model and generated predictions

Step 6: Output Generation – Stored in forecast_output.csv (Actual vs Predicted values)

5. Results

- Predictions generated successfully
- Stored in CSV file
- Evaluated using MAE and RMSE
- Model shows good accuracy

6. Workflow

Dataset → Preprocessing → Training → Save Model → Prediction → Output → Evaluation

7. Usefulness

- Demand forecasting
- Capacity planning
- Resource allocation

Benefits: Reduces cost, saves time, improves efficiency

8. Applications

- Cloud resource management
- Business forecasting
- Energy prediction
- Financial analysis

9. Conclusion

Model successfully deployed, predictions generated and evaluated. Ready for real-world use and future improvements.